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United States Patent Application Publication

20250283690

Kind Code

A1

Publication Date

September 11, 2025

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SIGHT MOUNT AND RELATED METHOD OF USE

Abstract

A weapon mount is provided to connect an accessory to a weapon adjacent a sight. The mount can include a main body, a forward projecting first arm and a locking plate. The main body can engage a rear of the sight, for example, a front sight post, and the locking plate can engage a front of the sight. The locking plate, main body and mount can be centered on the post via a centering fastener projecting through a center plate hole of the locking plate, and can be clamped to the sight post via first and second fasteners projecting through the locking plate and engaged with the first arm and a second arm which can be part of the main body or a modular body. The modular body can be slidably joined with the main body via a guide rail and corresponding slot. A related method is provided.

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Family ID: 1000008812388

Appl. No.: 19/097270

Filed: April 01, 2025

Related U.S. Application Data

parent US division 18600909 20240311 parent-grant-document US 12281873 child US 19097270

Publication Classification

Int. Cl.: F41C27/00 (20060101); F41G11/00 (20060101)

U.S. Cl.:

Background/Summary

BACKGROUND OF THE INVENTION

[0001] The present invention relates to a weapon mounting system, and more particularly to a firearm mount for mounting accessories to a weapon such as a firearm.

[0002] Many modern sporting and military firearms include a handguard that extends forward of a receiver, and generally around a barrel of the firearm. The handguard prevents contact between the user and the barrel, thereby protecting the user when the barrel heats up after extended periods of fire. From the manufacturer, many handguards on stock firearms have inadequate or inaccessible mounting locations. Some stock firearm handguards are void of rails or other surfaces upon which to mount accessories, such as lights, lasers, pressure switches and other items.

[0003] For example, in an MP5 firearm, manufactured by Heckler & Koch of Germany, or variants thereof, a tube extends above the barrel, away from the receiver of the firearm. A stock handguard over that tube is rather spartan and simple, without any decent accessory rails or mounts. Due to the front sight projecting upward adjacent the front of the stock handguard and mounted to the barrel, the handguard and tube have no good attachment points for accessories or other components. Thus, most handguards and rails for the MP5 are secured only to the receiver, or alternatively and separately under the tube. This reduces the amount of space for accessory mounting, particularly lights, lasers and pressure switches for the same.

[0004] Accordingly, there remains room for improvement in the field of mounting rails and handguards, and in particular, the way that they are secured to a firearm and their orientation relative to certain parts of the firearm for ease of access and actuation.

SUMMARY OF THE INVENTION

[0005] A weapon mount is provided to connect an accessory to a weapon adjacent a sight. The mount can include a main body, a forward projecting first arm and a locking plate. The main body can engage a rear of the sight, for example, a front sight post, and the locking plate can engage a front of the sight for securement of the mount to the sight.

[0006] In one embodiment, the locking plate, main body and mount can be centered on the post via a centering fastener projecting through a center plate hole of the locking plate and can be clamped to the sight post via first and second fasteners projecting through the locking plate and engaged with the first arm and a second arm which can be part of the main body and/or a modular body.

[0007] In another embodiment, the main body can include a front edge and the first arm can extend forwardly of the front edge. The locking plate can face toward the front edge and the sight of the weapon can be disposed between the front edge and the locking plate. The locking plate can be urged toward the main body to clamp the sight, which can be in the form of a sight post, between the main body and locking plate.

[0008] In still another embodiment, the locking plate can define a first hole through which a first fastener projects and a second hole through which a second fastener projects. The first fastener can join the locking plate to the first arm. The second fastener can join the locking plate to a second arm.

[0009] In yet another embodiment, the locking plate can define a center plate hole distal from the first hole and second hole, optionally between those holes. These holes can all be defined in a forward surface of the locking plate facing away from the sight post.

[0010] In even another embodiment, a modular body can be joined with the main body and the locking plate opposite the first arm, the modular body including or forming the second arm. A first rail can be joined with the main body, and a second rail joined with the modular body. The first rail

and the second rail can slidably join with one another, thereby joining the main body and the modular body.

[0011] In a further embodiment, the guide rail can be a dovetail rail, and the guide rail slot can be a dovetail recess that receives the dovetail rail.

[0012] In still a further embodiment, the main body can include an integrally formed second arm that extends parallel to the first arm and is configured to extend on an opposite lateral side of the sight post when the mount is installed relative thereto. The first arm and second arm can straddle the sight post, and the locking plate can be secured to the front ends of each of the arms with respective first and second fasteners.

[0013] In yet a further embodiment, a first elastomeric spacer can be disposed around the first fastener and a second elastomeric spacer disposed around the second fastener. The first elastomeric spacer is located between the locking plate and the first arm, and the second elastomeric spacer is located between the locking plate and the modular body. The elastomeric spacers can provide variability in the distance between the front edge of the main body and the rear face of the locking plate to accommodate sight posts of varying widths.

[0014] In even a further embodiment, the main body can include a plate that extends rearward from the sight post. The plate can include the connector. The connector can be a plurality of holes defined by the plate and configured to receive M-LOK, Key Mod or other type connectors of accessories to mount to the main body.

[0015] In another embodiment, the main body can include an upper mounting panel and a first side mounting panel extending laterally away from the upper mounting panel at an angle such as 45 degrees. The modular body can include a second mounting panel extending away from the upper mounting panel opposite the first side mounting panel at an angle, such as 45 degrees. The side panels can extend downward along sides of a tube which can house a cocking lever rod or slide of the weapon. Where included, certain connector holes can be located on both the upper panel and the 45 degree side panels, allowing for the attachment of lights, pressure switches and other accessories in a tactical setup.

[0016] In still another embodiment, the mount can facilitate mounting of an accessory, such as a light, a laser, a sling mount, a grip, a hand stop, a rail, a pressure switch or other accessories. The accessory can mount to a connector which can be in the form of a hole of varying shapes and sizes.

[0017] In still another embodiment, a method is provided. The method can include placing a main body including at least one connector configured to connect an accessory to the main body so that a front edge of the main body engages a rear surface of a front sight post of weapon, with a first arm extending forwardly of the front edge along a first lateral side of the sight post, a second arm extending forwardly of the front edge along a second lateral side of the sight post; placing a locking plate facing toward the front edge so that the sight post is sandwiched between the locking plate and the front edge, the locking plate defining a first hole through which a first fastener projects, the first fastener joining the locking plate to the first arm, the locking plate defining a center plate hole distal from the first hole, the locking plate defining a second hole through which a second fastener projects, the first hole, second hole and center plate hole facing forwardly of the sight post; and tightening a centering fastener in the center plate hole of the locking plate to engage a center sight hole defined by the sight post to center the main body relative to the sight post between the first lateral side and the second lateral side.

[0018] In yet another embodiment, the method can enable a user to connect the accessory to the mount while the main body, first arm, second arm and locking plate are disposed around and secured to the sight post holding the main body and at least one connector fixedly rearward of the sight post.

[0019] The current embodiments provide a weapon mount that joins easily and rigidly to a variety of firearms, including but not limited to an MP5 firearm or variants thereof. The mount can be directly attached to a front sight of a weapon, such as a firearm, without permanent modifications

to the weapon, and can be compatible with standard handguards on stock weapons. Where included, the center fastener can easily and consistently center the mount relative to the sight, for example a sight post, which in turn can center the main body relative to the centerline of the weapon and allowing precise alignment of an associated accessory. This can provide precise alignment of that mount and thus the accessory relative to the weapon. Where included, the locking plate can combine all fasteners used with the mount in a forward-facing orientation to ease assembly and quickly clamp and center the mount relative to the sight. A user thus need not tighten multiple screws from different locations and orientations relative to the mount. Where included, the elastomeric spacers can allow the mount to accommodate a variety of sights which might be differently sized or out of tolerance. Where included, the modular body can make assembly and securement of the mount to the sight post easier and more convenient. Such a modular body also can offer flexibility with modular design, including options for varying numbers of M-LOK slots, direct attachment of pressure switches and/or a two-piece modular concept.

[0020] These and other objects, advantages, and features of the invention will be more fully understood and appreciated by reference to the description of the current embodiment and the drawings.

[0021] Before the embodiments of the invention are explained in detail, it is to be understood that the invention is not limited to the details of operation or to the details of construction and the arrangement of the components set forth in the following description or illustrated in the drawings. The invention may be implemented in various other embodiments and of being practiced or being carried out in alternative ways not expressly disclosed herein. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including” and “comprising” and variations thereof is meant to encompass the items listed thereafter and equivalents thereof as well as additional items and equivalents thereof. Further, enumeration may be used in the description of various embodiments. Unless otherwise expressly stated, the use of enumeration should not be construed as limiting the invention to any specific order or number of components. Nor should the use of enumeration be construed as excluding from the scope of the invention any additional steps or components that might be combined with or into the enumerated steps or components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. 1 is a side view of the mount of a current embodiment secured to a weapon with an exemplary light accessory joined to the mount and thus to the weapon.

[0023] FIG. 2 is a perspective view of the mount on the weapon and mounted to a front sight post thereof.

[0024] FIG. 3 is a front view of the mount.

[0025] FIG. 4 is an exploded view of the mount showing a main body, an elongated modular body and a locking plate.

[0026] FIG. 5 is a perspective view of the mount showing the main body, a first alternative shorter modular body and a locking plate in assembled form.

[0027] FIG. 6 is a perspective view of the mount showing the main body, a second alternative rail modular body and a locking plate in assembled form.

[0028] FIG. 7 is a perspective view of a third alternative embodiment of the mount showing the main body and a locking plate in assembled form.

[0029] FIG. 8 is an exploded view of a fourth alternative embodiment of the mount showing a main body, an elongated modular body and a locking plate.

DETAILED DESCRIPTION OF THE CURRENT EMBODIMENTS

[0030] A current embodiment of the weapon mount is shown in FIGS. 1-3 and generally designated **10**. The weapon mount **10**, which can also be referred to as an accessory mount or mount in general, optionally is configured to be secured to a front sight **1** of a weapon **8**, such as a firearm. The weapon **8** can be a sporting, military or hunting rifle or pistol, for example an MP5 from Heckler & Koch of Germany, variants thereof and other firearm or weapon systems that include a barrel, receiver and an optional cocking lever housing and/or gas tube. The mount **10** can include a main body **20** that extends rearward of the front sight **1**, which can be a front sight post, and can include a first arm **21** that extends past a first lateral side **1L** forward of the main body **20**. A locking plate **30** can be secured forward of the front sight **1** to the first arm **21** with a first fastener **31**. The locking plate **30** can be secured to a second arm **42** with a second fastener **32** which optionally can be part of a modular body **40** or integral with the main body **10** itself as described below. The locking plate **30** can include a center plate hole **50** through which a centering fastener **52** is installed to engage a center sight hole **1C** defined by the sight **1** of the weapon to center the locking plate **30** and thus the mount **10** relative to the sight **1**.

[0031] Turning now to FIGS. 1-3, the firearm mount **10** and components of the weapon **8** will now be described in further detail. As mentioned above, the mount **10** mounts over and above the cocking lever housing **4**, as well as a barrel **2** and stock handguard of the firearm. The mount can be installed to the front sight **1**, which can be a front sight post including a main post **1P** that extends upward to a ghost ring or circle aperture **1G** above the main post **1P**. The mount can engage the main post **1P** below the ghost ring **1G**, optionally without engaging the handguard and/or cocking lever housing **4**, also referred to as a tube herein. The barrel **2** can include a muzzle end **2M** and a receiver end **2R**, with a longitudinal axis **BA** extending therebetween. The receiver end **2R** can extend from the receiver **6**. The cocking lever housing or tube **4** can extend forwardly from the receiver **6**. In some cases, that lever housing **4** can extend as a tube from a portion of the receiver **6** over the barrel **3**. The components of the housing **4** and receiver **6** can be stamped from metal parts or otherwise formed or machined. The cocking lever housing **4** can house a cocking lever **4C** that is designed to move a bolt within the receiver **4** to load a bullet in a chamber of the associated barrel **2**.

[0032] With further reference to FIGS. 1-4, the mount **10** can include the main body **20**. The main body **20** can include an upper panel **20U** and a first side mounting panel **20S** extending laterally away from the upper mounting panel at an angle **A1**, which optionally can be about 45 degrees or some other angle. The main body optionally can be joined with a modular body **40** as described below, which modular body **40** can include a second side panel **40S** extending laterally away from the upper panel at an angle **A2**, which optionally can be about 45 degrees or some other angle. The side panels can extend away from one another and the upper panel on opposite sides of the tube **4**. The configuration of the upper and side panels optionally can be such that they encase or at least partially surround an upper portion of the tube **4** of the weapon rearward of the sight **1**.

[0033] Optionally, as shown in FIG. 4, the main body can include an interior surface or contour **201**. This interior contour can be arcuate or curvilinear as shown there. The interior surface can extend under the upper panel and through the side panel. The interior surface, where arcuate, can fit around or generally accommodate the tube **4** of the weapon rearward of the sight. The modular body **40** optionally can include a similarly arcuate or curvilinear contour to fit around the tube **4**. Of course, the interior contour **201** can be angular in other applications to fit around the tube with more spacing between the tube and the mount.

[0034] The main body **20** shown in FIGS. 2 and 4 can include one or more connectors **25** which are configured to connect an accessory to the main body. The one or more connectors can be one or more holes, recesses, slots or other apertures defined by the panels or plates. The connectors can be in the form of elongated slots with rounded ends. The connectors can be configured to receive M-LOK, Key Mod or other type connectors of accessories to mount the same to the main body. The connectors can be aligned with the length of the panels on the various sides of the main body and

modular body where included.

[0035] The accessory **90** can be a light as shown, a laser, a switch, a grip, a stop or any other type of accessory. The accessory also can be actuated by a pressure switch **92**. The light **90** can be attached to the modular body **40**, while the pressure switch or other type of actuator or switch **92** can be joined with the main body. More particularly, the switch can be joined with the side panel **40S** at the angle **A2** while the pressure switch **92** can be joined with the upper panel **20U** atop the main body and behind the sight **1**. The light itself can extend forwardly past a portion of the sight **1**.

[0036] The main body **20**, as mentioned above, can include a first engagement surface **23** configured to engage a first sight surface **1R** of the sight **1**, which can be at the rear of the sight post **1P**. The first engagement surface **23** can be a front edge of the upper panel **20U** or some other forward edge or portion of the main body. The front edge can face forwardly toward the muzzle **2M**, and can be flat or planar depending on the application. The main body can include a first arm **21** that can extend forwardly beyond the main body optionally adjacent and along a first lateral side **1L** of the sight **1**, beyond the first engagement surface **23** or front edge. The first sight surface **1R** can be transverse and/or perpendicular to a first lateral side **1L** and second lateral side **1R** of the sight. The first sight surface **1R** can be parallel to a second sight surface which can be a forward surface of the sight or sight post **1P**.

[0037] As shown in FIG. **4**, the first arm **21** can be integrally formed with the main body **20** as a single piece unit. The first arm can be of a smaller width **WA** than the width **WS** of the side panel **20S**. The arm **21** can include a flat or planar inner side **211** that faces the first lateral side **1L**. The arm can extend parallel to the barrel axis **BA** when the mount is installed and can terminate at a distal end **21E**. The distal end **21E** can define a recess **21R** bounded by a shoulder **21S** along an outer periphery thereof. A first arm fastener hole **21H** can be defined and optionally centered within the recess **21R**. The recess **21R** optionally can be configured to receive an elastomeric spacer or washer **38** that can be seated or sandwiched between the locking plate **30** and the arm **21**. The elastomeric spacer optionally can be an O-ring or washer constructed from rubber, silicone, polymers or other elastomeric materials. Optionally, a second arm **42** of the module body can define a similar recess **42R** and shoulder **42S**, along with a second arm fastener hole **42H**.

[0038] The first elastomeric spacer **38** and a second elastomeric spacer **39** described below can be compressed within the recess between the locking plate and the arms when the locking plate is secured with fasteners. For example, as shown in FIG. **2**, the first elastomeric spacer **38** can be disposed around the first fastener **31**. The second elastomeric spacer **39** can be disposed around the second fastener **32**. The first elastomeric spacer can be located between the locking plate **30** and the first arm **21**, and the second elastomeric spacer **39** can be located between the locking plate **30** and the modular body **40**.

[0039] Further optionally, in some applications, the arm can include a locating counter (not shown) to receive a portion or lobe of the plate to align the locking plate with the arm, and more particularly to align the first fastener hole **31H** defined by the plate with the first arm fastener hole **21H** to facilitate installation of the first fastener **31** into those holes to attach the plate to the main body around the sight. The first arm fastener hole **21H** can be defined within the recess **21R** distal and offset from the contour, which can project beyond the recess and around at least a portion of the hole **21H**. The contour in some cases can be a pin (not shown) that fits in a corresponding pin hole in the locking plate to facilitate alignment. Other alignment mechanisms are contemplated for the fastener holes and fasteners.

[0040] With reference to FIGS. **2-4**, the locking plate **30** can face toward the front edge **23** of the main body. The plate can be configured so the sight **1** and in particular the main post **1P** of the sight is disposed between the front edge and the locking plate, so that when the fasteners **31** and **32** are tightened, the main post **1P** is clamped between the front edge and locking plate. The locking plate **30** can define a first hole **31H** through which a first fastener **31** projects. The first hole optionally is unthreaded. That first fastener **31** can extend through the locking plate, and the spacer where

included, and into the hole **21H**, which optionally can be threaded. The first fastener when disposed through the plate and into the hole **21H** can join the locking plate **30** to the first arm **21**. The locking plate **30** can define a second hole **32H** through which a second fastener **32** projects. The second hole optionally is unthreaded. That second fastener **32** can extend through the locking plate, and the spacer where included, and into the hole **42H**, which optionally can be threaded. The second fastener when disposed through the plate and into the hole **42H** can join the locking plate **30** to the second arm **42**.

[0041] Optionally, the fasteners **31** and **32** can include heads. The plate **30** can define head recesses to accommodate the heads. Further optionally, as shown in FIGS. **2** and **4**, the fastener holes **31H**, **32H** and the fasteners **31**, **32** can have longitudinal fastener axes FA that are parallel to the barrel axis BA. The locking plate and front surface optionally can be perpendicular to the longitudinal axis BA of the barrel.

[0042] The locking plate **30** also can define a center plate hole **50** distal from the first hole **31H** and second hole **32H**. The first hole, second hole and center plate hole can be defined in a forward surface **30F** of the locking plate **30**. The center plate hole can be defined midway between the first hole and the second hole. The center plate hole, first hole and second hole optionally can be defined in a common plane P1 extending through the locking plate rearward of the forward surface **30F**. The center plate hole can be threaded to receive a centering fastener **52**. The centering fastener can be a screw, for example a set screw, optionally including a tapered or conical point **52T**. The point can be configured to engage a center sight hole **1C** defined by the sight **1** and in particular at the center of the sight post **1P** between the first **1L** and second **1R** lateral sides of the sight. This center sight hole **1C** can be formed by the original equipment manufacturer of the sight and sight post, and optionally the weapon. This center sight hole **1C** can be defined in the side to side center of the second sight surface **1F** defined at the front of the sight **1** and optionally on the front of the main post **1P**. Upon this engagement and tightening of the fastener **52**, the locking plate **30** and the attached main body can be consistently and precisely centered on the sight **1** and in particular the main sight post **1P**.

[0043] With reference to FIGS. **2** and **4**, the mount **10** can include the modular body **40** as described above. This modular body can include the side panel **40S** with corresponding connectors **45**. An accessory, such as a light **90** or other accessory, can be fastened to the modular body via these connectors. The modular body and thus any attached accessory thereto therefore can be selectively attachable to and removable from the main body and locking plate via the above noted second fastener **32**, and perhaps another feature as described below. As with the first fastener, the second fastener **32** can project through the faster hole **32H** and into the second arm fastener hole **42H** into which it can be threaded. The elastomeric spacer **39** can be disposed there between and optionally within a recess **42R** defined by the distal end of the second arm **42**. The modular body and the second arm **42** can extend generally in parallel to the first arm **21** as well as the barrel axis BA. The second arm **42** can be configured for positioning adjacent the second lateral side surface **1R** opposite the first lateral side surface **1L**. Optionally, like the first arm, the interior surface of the second arm may or may not engage the respective lateral side surface **1R** of the sight **1**.

[0044] As shown in FIG. **2**, when the mount **10** is mounted to the front sight **1**, the forward edge **23**, the first arm **21**, the second arm **42** and the locking plate **30**, generally surround or are disposed around that front sight and the main post **1P** of the sight. When the fasteners **31** and **32** are tightened, they bring the rear surface of the locking plate toward the forward ends of the first and second. Where included, the elastomeric spacers are compressed between the locking plate and the ends of the arms, optionally compressing within the respective recesses defined by the ends of the arms. The centering fastener **52** previously can be tightened to center the locking plate and main body relative to the sight post, which again can be rigidly and immovably attached to the tube and/or barrel of the firearm. Upon this compression of the sight post between the locking plate and the main body, the mount effectively clamps onto the sight post in a fixed and rigid configuration

and orientation. This optionally is accomplished without modifying the handguard 7H, the tube 4 or other components of the weapon, and further optionally via only interaction of the mount with the front sight post and at most perhaps the tube 4 via the interior of the mount contacting it in some applications. Thus, no modification or attachment to the handguard of other elements besides the sight is used to attach the mount and thus the accessory to the weapon.

[0045] As mentioned above, the modular body 40 can be removably attached to the main body and/or the locking plate, and thus, so can any accessory be attached thereto for easy replacement or change-out. The second fastener 32 can be used to secure the second arm 42 and thus the modular body to the locking plate. The modular body 40 however, can include an interlocking feature 46 that is removably joinable and interfitting with a corresponding interlocking feature 26 of the main body. Optionally, as shown in FIG. 4, the first interlocking feature 26 can be a first rail joined with the main body 20. The second interlocking feature 46 can be a second rail joined with or defined by the modular body. The first rail and second rail can be slidably joined with one another such that the first rail can be slid relative to the second rail and vice versa. Although shown with the second rail 46 being defined as a recess or slot on the modular body, and the first rail 26 being defined as a projection on the main body, these components can be reversed such that the recess is defined in the main body and the projection is included on the modular body.

[0046] Optionally, as shown in FIG. 4, the first rail 26 can be in the form of a dovetail rail. The second rail 46 can be in the form of a dovetail recess or slot that receives the dovetail rail 26. Of course, other shapes of these rails and recesses can be used to movably connect and/or interlock the modular body relative to the main body. Indeed, the interlocking features may not even be rails and instead can be fasteners, cams, clamps or other movable components.

[0047] Further optionally, the interlocking rails or other features can be set up to provide a fixed amount of movement or sliding relative to one another. For example, the main body can include a shoulder or stop 28 joined thereto, adjacent an opening 200 which can be defined between the front edge 23, the arm 21 and the locking plate 30. A corresponding shoulder 48 can be included on the modular body 40, rearward or adjacent the second arm 42. The shoulder 48 can engage the stop 28 when the modular body 40 is slid or moved far enough forward relative to the main body 20 and the second arm 42 optionally projects forward of the forward edge 23 the same distance D1 as the first arm 21. Of course, other stops and limiters can be utilized to provide precise movement and positioning of the modular body relative to the other components.

[0048] A method of using the weapon mount 10 of the current embodiment will now be described with reference to FIGS. 1-4. Generally, the method can comprise placing a main body including at least one connector configured to connect an accessory to the main body so that a front edge of the main body engages a rear surface of a front sight post. A first arm can extend forwardly of the front edge along a first lateral side of the sight post. A second arm can extend forwardly of the front edge along a second lateral side of the sight post. The method can include placing a locking plate facing toward the front edge so that the sight post is sandwiched between the locking plate and the front edge. The locking plate can define a first hole through which a first fastener projects. The first fastener can join the locking plate to the first arm. The locking plate can define a center plate hole distal from the first hole. The locking plate can define a second hole through which a second fastener projects.

[0049] Optionally, the first hole, second hole and center plate hole can face forwardly of the sight post, and can be aligned with the barrel axis. The method can include tightening a centering fastener in the center plate hole of the locking plate to engage a center sight hole defined by the sight post to center the main body relative to the sight post between the first lateral side and the second lateral side. A user can connect the accessory to the mount while the main body, first arm, second arm and locking plate are disposed around and secured to the sight post holding the main body and at least one connector fixedly rearward, forward and/or adjacent the sight post.

[0050] The method can include tightening the first fastener to compress a first elastomeric spacer

between the locking plate and the first arm and tightening the second fastener to compress a second elastomeric spacer between the locking plate and the second arm. The method can include sliding the second arm relative to the main body such that a guide rail moves relative to a guide rail slot. The method can include aligning the first fastener, the second fastener and the centering fastener so that each are parallel in length to one another.

[0051] A first alternative embodiment of the mount is shown in FIG. 5 and generally designated **110**. This mount can be similar to that of the current embodiment in structure, function and operation with several exceptions. For example, the mount **110** can include the main body **120** and a locking plate **130** secured thereto with fasteners, which generally clamps the main post **1P** of a front sight **1**. In this embodiment, however, the modular body **140** only includes a single connector **145** in the form of a single slot. The modular body and its side panel **140S** also are a shorter length **L2** than the length **L1** of the modular body **40** shown in FIG. 2. This in turn can reduce the overall size of the modular body and the mount **110**, yet still provide adequate area to mount an accessory via the mouth **110** to a weapon.

[0052] A second alternative embodiment of the mount is shown in FIG. 6 and generally designated **210**. This mount can be similar to that of the current embodiment in structure, function and operation with several exceptions. For example, the mount **210** can include the main body **220** and a locking plate **230** secured thereto with fasteners, which generally clamp to the main post **1P** of a front sight **1**. In this embodiment, however, the modular body **240** can be in the form of a picatinny rail **246** that extends from the main side panel **240S**. One or more additional connector holes **245** can be defined in the picatinny rail to assist in further connecting or mounting an accessory to the modular body **240**. This modular body **240** also can include a length **L2** that is less than the length **L1** of the modular body **40** of the embodiment above.

[0053] A third alternative embodiment of the mount is shown in FIG. 7 and generally designated **310**. This mount can be similar to that of the current embodiment in structure, function and operation with several exceptions. For example, the mount **310** can include the main body **320** and a locking plate **330** secured thereto with fasteners, which generally clamp to the main post **1P** of a front sight **1**. In this embodiment, however, the modular body is absent, and the main body **320** includes an integral side panel **340S**. The second arm **342** also can be integrally formed with the main body, along with the first arm as a single piece unit.

[0054] A fourth alternative embodiment of the mount is shown in FIG. 8 and generally designated **410**. This mount can be similar to that of the current embodiment in structure, function and operation with several exceptions. For example, the mount **410** can include the main body **420** and a locking plate **430** secured thereto with fasteners, which generally clamp to the main post **1P** of a front sight **1**. In this embodiment, however, the main body **420** can be shorter than the main body **20** in the current embodiment. For example, the respective upper panel **420U** and side panel **420S** can be a length **L3** that is substantially shorter than the length of those panels **20U** and **20S** in the current embodiment. These panels also can define fewer connectors, for example only one connector. In this embodiment, the modular body **440** can be larger than the modular body of the current embodiment **40** described above. It also can include an upper panel **440U** that attaches to a side panel **440S** to increase the overall mounting area of the modular body. Optionally, this modular body **440** can be shortened at the line **LA** so that the overall size of the mount is compact. Of course, the modular body and main body can be changed in size and shape, as well as connecting and interlocking with one another depending on the application.

[0055] Although the different elements and assemblies of the embodiments are described herein as having certain functional characteristics, each element and/or its relation to other elements can be depicted or oriented in a variety of different aesthetic configurations, which support the ornamental and aesthetic aspects of the same. Simply because an apparatus, element or assembly of one or more of elements is described herein as having a function does not mean its orientation, layout or configuration is not purely aesthetic and ornamental in nature.

[0056] Directional terms, such as “vertical,” “horizontal,” “top,” “bottom,” “upper,” “lower,” “inner,” “inwardly,” “outer” and “outwardly,” are used to assist in describing the invention based on the orientation of the embodiments shown in the illustrations. The use of directional terms should not be interpreted to limit the invention to any specific orientation(s).

[0057] In addition, when a component, part or layer is referred to as being “joined with,” “on,” “engaged with,” “adhered to,” “secured to,” or “coupled to” another component, part or layer, it may be directly joined with, on, engaged with, adhered to, secured to, or coupled to the other component, part or layer, or any number of intervening components, parts or layers may be present. In contrast, when an element is referred to as being “directly joined with,” “directly on,” “directly engaged with,” “directly adhered to,” “directly secured to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between components, layers and parts should be interpreted in a like manner, such as “adjacent” versus “directly adjacent” and similar words. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0058] The above description is that of current embodiments of the invention. Various alterations and changes can be made without departing from the broader aspects of the invention as defined in the appended claims, which are to be interpreted in accordance with the principles of patent law including the doctrine of equivalents. This disclosure is presented for illustrative purposes and should not be interpreted as an exhaustive description of all embodiments of the invention or to limit the scope of the claims to the specific elements illustrated or described in connection with these embodiments. For example, and without limitation, any individual element(s) of the described invention may be replaced by alternative elements that provide substantially similar functionality or otherwise provide adequate operation. This includes, for example, presently known alternative elements, such as those that might be currently known to one skilled in the art, and alternative elements that may be developed in the future, such as those that one skilled in the art might, upon development, recognize as an alternative. Further, the disclosed embodiments include a plurality of features that are described in concert and that might cooperatively provide a collection of benefits. The present invention is not limited to only those embodiments that include all of these features or that provide all of the stated benefits, except to the extent otherwise expressly set forth in the issued claims. Any reference to claim elements in the singular, for example, using the articles “a,” “an,” “the” or “said,” is not to be construed as limiting the element to the singular. Any reference to claim elements as “at least one of X, Y and Z” is meant to include any one of X, Y or Z individually, any combination of X, Y and Z, for example, X, Y, Z; X, Y; X, Z; Y, Z, and/or any other possible combination together or alone of those elements, noting that the same is open ended and can include other elements.

[0059] Reference throughout this specification to “a current embodiment” or “an embodiment” or “alternative embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment herein. Accordingly, the appearance of the phrases “in one embodiment” or “in an embodiment” or “in an alternative embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner with or in one or more embodiments.

Claims

1. A weapon mount method comprising: placing a main body including at least one connector configured to connect an accessory to the main body so that a front edge of the main body is configured to engage a rear surface of a front sight post of weapon, with a first arm configured to extend forwardly of the front edge along a first lateral side of the sight post, a second arm configured to extend forwardly of the front edge along a second lateral side of the sight post;

placing a locking plate facing so that the locking plate is configured to face toward the front edge such that the locking plate and the front edge are configured to sandwich the sight post therebetween, the locking plate defining a first hole through which a first fastener projects, the first fastener joining the locking plate to the first arm, the locking plate defining a center plate hole distal from the first hole, the locking plate defining a second hole through which a second fastener projects, the first hole, second hole and center plate hole configured to face forwardly of the sight post; and arranging a centering fastener in the center plate hole of the locking plate so that the centering fastener is configured to engage a center sight hole defined by the sight post whereby the main body is able to be centered relative to the sight post between the first lateral side and the second lateral side, whereby when installed relative to the sight post, a user can connect the accessory to the mount while the main body, first arm, second arm and locking plate are disposed around and secured to the sight post holding the main body and at least one connector fixedly rearward of the sight post.

2. The method of claim 1 comprising: tightening the first fastener to compress a first elastomeric spacer between the locking plate and the first arm; and tightening the second fastener to compress a second elastomeric spacer between the locking plate and the second arm.

3. The method of claim 1 comprising: sliding the second arm relative to the main body such that a guide rail moves relative to a guide rail slot.

4. The method of claim 1 comprising: aligning the first fastener, the second fastener and the centering fastener so that each are parallel in length to one another.

5. The method of claim 1, comprising: slidably joining a first rail with a second rail, wherein the first rail is on the main body; wherein the second rail is on a modular body.

6. The method of claim 5, wherein the first rail and the second rail include corresponding, interfitting elements, wherein the first rail and the second rail terminate short of the first arm and the second arm.

7. The method of claim 6, wherein the corresponding, interfitting elements are dovetail shaped.

8. The method of claim 1 comprising: nesting a guide rail in a guide rail slot; and sliding the guide rail in the guide rail slot while moving the second arm along a side surface of the sight.

9. The method of claim 8 comprising: aligning a forward portion of the second arm with a forward surface of the sight post; and attaching the lock plate to the forward portion of the second arm by installing the second fastener in a threaded hole defined by the second arm.

10. The method of claim 1 comprising: engaging the main body against the rear surface of a front sight post; engaging at least one of the lock plate and the centering fastener against a front surface of the sight post.

11. A weapon mount method comprising: positioning a main body including at least one connector configured to connect an accessory to the main body, a front edge, and a first arm extending forwardly of the front edge; facing a locking plate toward the front edge so a sight of the weapon is disposable between the front edge and the locking plate, the locking plate defining a first hole through which a first fastener projects, the first fastener joining the locking plate to the first arm, the locking plate defining a center plate hole distal from the first hole, the locking plate defining a second hole through which a second fastener projects, the first hole, the second hole and the center plate hole defined in a forward surface of the locking plate; and arranging a centering fastener in the center plate hole of the locking plate so the centering fastener is configured to engage a center sight hole defined by the sight, whereby the main body, the first arm and the locking plate are disposable around the sight for securement of the mount to the sight.

12. The method of claim 11 comprising: positioning a modular body along the main body and distal from the first arm, so that the modular body extends parallel to the first arm.

13. The method of claim 11, comprising: placing a first elastomeric spacer around the first fastener and adjacent the locking plate; and placing a second elastomeric spacer around the second fastener and adjacent the locking plate.

- 14.** The method of claim 12, slidably nesting a guide rail and a guide rail slot with one another to secure the main body to a modular body.
- 15.** The method of claim 11 comprising: surrounding the sight post with the main body front edge, the first arm, the locking plate; and a second arm opposite the first arm; and fastening the locking plate to the first arm with the first fastener to secure the main body, the first arm, the locking plate and the second arm around the sight post in a fixed and immovable manner.
- 16.** The method of claim 11 comprising: placing a first lateral wing and a second lateral wing of the main body so that the first lateral wing and the second lateral wing extend laterally away from one another on opposite sides of a tube of the weapon.
- 17.** The method of claim 11 comprising: securing an accessory to the first lateral wing but not the second lateral wing.
- 18.** The method of claim 11 comprising: interlocking a modular body with the main body via a guide rail such that the second arm extends forward of the front edge of the main body.
- 19.** The method of claim 12 comprising: installing a second fastener distal from the first fastener relative to a second arm to secure the locking plate to the second arm.
- 20.** A weapon mount method comprising: positioning a main body including at least one connector configured to connect an accessory to the main body and a front edge; extending a first arm and a second arm forwardly of the front edge, the first arm and second arm joined with a locking plate that faces toward the front edge so a sight of the weapon is disposable between the front edge and the locking plate, the locking plate defining a first hole in which a first fastener is disposed, the locking plate defining a center plate hole distal from the first hole, the locking plate defining a second hole in which a second fastener is disposed, the first hole, second hole and center plate hole defined in a forward surface of the locking plate; and arranging a centering fastener in the center plate hole of the locking plate so the centering fastener is configured to engage a center sight hole defined by the sight, whereby the main body, the first arm and the locking plate are disposable around the sight for securement of the mount to the sight.
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